QA Tester Occupancy

Calculation guide with worked examples

The dashboard converts each current assignment into weighted capacity points. Eight points equal 100% occupancy. Completed requests are shown for tracking but do not contribute to occupancy.

1. The formula

Occupancy % = round((total current assignment points ÷ 8) × 100)

- Only requests in a current tester/analyst workload status contribute points.
- Terminal/completed requests appear in request tracking, but add zero occupancy points.
- Functional and Performance requests assigned to multiple testers divide their points equally between those testers.
- SAST and DAST requests have one assigned Security Analyst, so their points are not divided.
- The displayed percentage is rounded to the nearest whole number by the application.

2. Capacity bands

Displayed occupancy	Band	Meaning
0%	Available	No weighted current assignments.
1–49%	Light	Capacity is available.
50–79%	Balanced	Meaningful active workload, below the high threshold.
80–99%	High	Approaching planned capacity.
100%	Full	At planned capacity: 8 points.
>100%	Overloaded	Assignments exceed the planned 8-point capacity.

3. Assignment weights

Functional testing

Status	Points
Tester Assigned	0.5
Test Design	1.0
Execution in Progress	1.0
Defect Raised	0.5
Waiting for Fix	0.0
Retesting	0.75
QA Completed	0.15
QA Sign-off Pending	0.1
QA Signed Off	0.1
Requester Verification	0.05

Performance testing

Status	Points
Environment Setup	1.0
Script Development	1.0
Baseline	0.75
Load Test Execution	1.0
Result Analysis	0.25
Defect/Fix/Retest	0.75
Report	0.15
Sign-off Pending	0.1

SAST and DAST

Status	Points
Configuration	0.75

Scanning	1.0
Finding Validation	0.75
Remediation	0.5
Assigned to Requester	0.1
Waiting for Fix	0.0
Assigned to Lead	0.1
Rescan	0.75
Security Complete	0.15
Report Ready	0.1

4. Worked examples

Example 1 — One active Functional request

Assignment	Calculation	Points
Functional: Test Design	1.00	1.00

Total points: 1.00

Occupancy = (1.00 ÷ 8) × 100 = 12%

The exact value is 12.5%; the application displays a whole-number percentage.

Example 2 — Mixed Functional and Performance work

Assignment	Calculation	Points
Functional: Execution in Progress	1.00	1.00
Functional: Retesting	.75	.75
Performance: Baseline	.75	.75
Performance: Result Analysis	.25	.25

Total points: 2.75

Occupancy = (2.75 ÷ 8) × 100 = 34%

Different lifecycle stages represent different levels of active effort.

Example 3 — Shared request between two testers

Assignment	Calculation	Points
Functional: Test Design	1.00 ÷ 2 testers	.50

Total points: .50

Occupancy = (.50 ÷ 8) × 100 = 6%

Each tester receives half of the request's point value. The same division rule applies to shared Performance requests.

Example 4 — Waiting work does not consume capacity

Assignment	Calculation	Points
Functional: Waiting for Fix	0.00	0.00
SAST: Waiting for Fix	0.00	0.00
DAST: Scanning	1.00	1.00

Total points: 1.00

Occupancy = (1.00 ÷ 8) × 100 = 12%

Waiting requests remain visible in the workload counts, but their point contribution is zero.

Example 5 — Near-complete work

Assignment	Calculation	Points
Functional: QA Completed	.15	.15
Performance: Sign-off Pending	.10	.10
SAST: Report Ready	.10	.10

Total points: .35

Occupancy = (.35 ÷ 8) × 100 = 4%

Near-complete work retains a small capacity weight because follow-up may still be required.

Example 6 — Full capacity

Assignment	Calculation	Points

Eight current requests at 1.00 point each	8 × 1.00	8.00

Total points: 8.00

Occupancy = (8.00 ÷ 8) × 100 = 100%

Eight full-weight concurrent assignments equal the configured planned capacity.

Example 7 — Overloaded tester

Assignment	Calculation	Points
Ten current requests at 1.00 point each	10 × 1.00	10.00

Total points: 10.00

Occupancy = (10.00 ÷ 8) × 100 = 125%

Occupancy may exceed 100%. The dashboard labels this tester Overloaded rather than capping the calculation.

5. How the date filter behaves

- The period filter applies to completed-work history: All time, Last 3 days, Last 15 days, Last month, or custom From/To dates.
- Current assignments always remain visible, even when their request was created before the selected period.
- Changing the date filter does not change current occupancy; it changes which completed requests appear in the tester's request ledger.

6. Reading the dashboard correctly

Dashboard value	What it means
Assigned	Number of current requests attributed to the tester/analyst.
Active Work	Current assignments not classified as queued, waiting, or near complete.
Queued	Functional requests at Tester Assigned.
Waiting / Blocked	Current requests waiting on fixes or assigned back to a requester.
Near Complete	Current requests in completion, report, sign-off, or requester-verification stages.
Available Capacity	100 minus occupancy, never shown below 0%.
Request tracking	Current assignments plus completed requests in the selected historical period.

7. Important interpretation notes

- Occupancy is a planning indicator, not a timesheet and not a measure of individual performance.
- The weights estimate concurrent workflow demand; they do not represent hours worked.
- A tester can have many low-weight near-complete items without being equivalent to the same number of active execution assignments.
- QA Leads should review the underlying request ledger before making assignment decisions.

Calculation source: backend/app/routers/dashboard.py — TESTER_CAPACITY_POINTS and the Functional, Performance, and Security load maps. This guide reflects the implementation baseline dated 13 August 2026.